ECHO: The Signal That Was Never Sent

Case Report — Entangled Minds: The Quantathon

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GitHub Repository: https://github.com/areeshafatima4790-tech/ECHO-Quantum-Investigation

1. Introduction

The ECHO investigation explores whether quantum entanglement could explain unusual correlations between distant observers and whether these correlations can transmit information faster than light.

Quantum computing uses qubits, which can exist in superpositions of states. Entangled qubits can produce strong correlations when measured. However, entanglement cannot be used to send controllable messages faster than light. The purpose of this investigation is to examine the experimental clues, test possible explanations through quantum circuit simulations, and determine whether the results support the claim of faster-than-light communication.

2. Experiments and Results

Experiment 1: The Gatekeeper

The first experiment tested the X and Hadamard gates. The X gate flips (|0\rangle) to (|1\rangle), while the Hadamard gate creates a superposition with approximately equal probabilities of measuring 0 or 1. Applying the Hadamard gate twice returns the qubit to its original state. These results demonstrate basic quantum gate behavior but do not prove faster-than-light communication.

Experiment 2: The Honest Coin

The second experiment applied an (R_y(\theta)) rotation to a qubit. For (\theta=1.159) radians, the probability of measuring 1 is approximately 30%, while the probability of measuring 0 is approximately 70%. This explains the biased output mathematically. The bias alone does not prove tampering; circuit settings and execution records must be checked.

Experiment 3: The Two Observers

The third experiment investigated a Bell pair in the state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Measurements in the Z basis produce correlated outcomes, 00 or 11. Since classical systems can also produce matching results, the experiment additionally compares measurements in the X basis. The Bell pair retains correlations that distinguish it from a simple classical mixture of 00 and 11. However, these correlations do not allow controllable faster-than-light communication.

Experiment 4: The Damaged Message

The fourth experiment tested classical majority-vote error correction. Each bit is repeated three times, allowing correction when at most one bit in a group is flipped. For example, 010 decodes to 0, but 110 incorrectly decodes to 1. Therefore, the method can correct limited errors but cannot guarantee perfect recovery.

3. Evidence and Clues

The investigation revealed three important clues:

- **Biased measurement results:** The Honest Coin experiment produced an expected 70:30 distribution rather than 50:50. This suggests that the circuit settings must be examined, but it does not independently prove tampering.
- **Correlated observations:** The Two Observers experiment showed correlated Bell-pair measurements. Comparing different measurement bases helps distinguish entanglement from a simple classical explanation. However, these correlations cannot transmit controllable messages faster than light.
- **Damaged bitstream:** The Damaged Message experiment demonstrated how corrupted bits can affect a received message and how majority voting can correct limited errors. This points to the importance of checking data integrity.

These clues help explain the experimental behavior, but they do not independently identify the cause of the incident or establish that anyone deliberately manipulated the system.

4. Interpretation and Alternative Explanation

The experiments demonstrate predictable quantum gate behavior, measurement bias, entanglement correlations, and classical error correction. However, they do not establish that ECHO transmitted information faster than light.

One explanation is that quantum correlations were mistaken for communication. The biased output could also result from circuit parameters rather than deliberate manipulation. Incomplete configuration records or machine-access logs are other possibilities. The simulations alone cannot determine the exact cause or identify anyone responsible, so accusations would be unjustified without independent evidence.

5. Limitations and Next Experiment

The investigation relies on simulations rather than complete original experimental records. Its results cannot independently establish what happened during the incident.

The next step is to verify notebook outputs, inspect circuit settings and random seeds, review machine-access logs, and investigate missing discrepancy reports. A more rigorous Bell test should use multiple measurement settings, independent records, appropriate controls, and statistical analysis.

6. Conclusion

The ECHO investigation demonstrates important quantum-computing principles but provides no evidence of controllable faster-than-light messaging. Quantum correlations are not equivalent to a communication channel. Further verification of experimental records is necessary before drawing conclusions about the incident.

7. References and AI Use

1. *ECHO Participant Manual*, supplied project material for Entangled Minds: The Quantathon.
2. IBM Quantum, *Qiskit Documentation*. https://docs.quantum.ibm.com/

3. **AI disclosure:** ChatGPT (OpenAI) assisted with explaining quantum concepts, reviewing debugging issues, organizing findings, and drafting this report. The team is responsible for verifying the results and understanding the code.